

EIGRP

Enhanced Interior Gateway Routing Protocol

Advance Distance Vector Protocol

It contains 3 table

- * Neighbor table Next Router Info
- * Topology table Best Route & Backup Route
- * Routing table Best Route

metric = Constant Parameter / K-values

K₁ = Bandwidth

K₂ = Load

K₃ = Delay

K₄ = Reliability

K₅ = MTU

Parameter	K1 Bandwidth	K2 Load	K3 Delay	K4 K5 Reliability
Formula	$K1 = BW$	$K2 = LM$ $256 / (256 - LM)$	$K3 = DLY$	$K4 = \frac{256}{256 - RLY}$
Default Value	1	1	1	1
Range	1 to 255	1 to 255	1 to 255	1 to 255

Default K-values

K₁ = 1

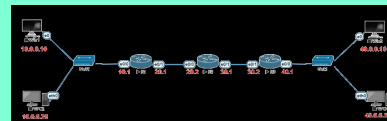
K₂ = 0

K₃ = 1

K₄ = 0

K₅ = 0

$$256 \left\{ \frac{10^7}{\text{Least bandwidth in Route}} + \text{total Delay} \right\}$$



Neighbor SHIP

By Sending Hello Packet automatically

RIP₁ 255.255.255.255

RIP₂ 224.0.0.9

Hello Packet
 SIP: 224.0.0.10
 DIP: 224.0.0.10
 EIGRP multicast IP

Neighbor condition:

Same network

Same AS

Same K-values

EIGRP timer

Hello time = 5 Sec

Hold time = 15 Sec



$$256 \left\{ \frac{10^7}{\text{Lost BW in Route}} + \text{total Delay} \right\}$$

$$256 \left\{ \frac{10^7}{10,000} + 300 \right\}$$

Thank-u
😊

$$256 \left\{ 1000 + 300 \right\}$$

$$256 \left\{ 1300 \right\}$$

$$= 332800$$

